



2011

YEAR 11

HSC ASSESSMENT TASK 1

Mathematics Extension 2

Date: Monday 5th December

Weighting: 10% of the HSC assessment: 5% Concepts, Skills and Techniques
5% Reasoning and Communication

General Instructions

- Reading time – 2 minutes
- Working time – 45 minutes
- Write using blue or black pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- All necessary working should be shown in every question

Total marks – 34

- Attempt all questions

Outcomes to be assessed:

- E2** A student chooses appropriate strategies to construct arguments and proofs in both concrete and abstract settings
- E3** A student uses the relationship between algebraic and geometric representations of complex numbers
- E4** A student uses efficient techniques for the algebraic manipulation required in dealing with questions such as those involving polynomials
- E9** A student communicates abstract ideas and relationships using appropriate notation and logical argument

Section I

Student Number: _____

4 marks

Attempt Questions 1 – 4

Circle the correct response (A, B, C or D) on the questions below.

1. Which of the following represents $z\bar{z}$?
- (A) $x^2 - y^2$ (B) $x^2 + y^2$ (C) 1 (D) $y^2 - x^2$
2. Which of the following represents $z + iw$, given $z = 3 - 4i$ and $w = 2 + i$.
- (A) $1 - 5i$ (B) $2 - 2i$ (C) $4 - 2i$ (D) $5 - 3i$
3. Which of the following represents $\frac{z}{w}$, given $z = 2 + 3i$ and $w = 3 - 2i$.
- (A) $\frac{12}{5} + \frac{13i}{5}$
- (B) $\frac{12}{13} + i$
- (C) $\frac{13i}{5}$
- (D) i
4. What is the general solution of $\tan 4\theta = 1$?
- (A) $\frac{\pi}{16} + \frac{n\pi}{4}$, where n is an integer.
- (B) $\frac{\pi}{16} + \frac{n\pi}{2}$, where n is an integer.
- (C) $\frac{\pi}{4} + \frac{n\pi}{4}$, where n is an integer.
- (D) $\frac{\pi}{4} + \frac{n\pi}{2}$, where n is an integer.

Section II

30 marks

Attempt questions 5 – 6

Answer each question on the paper provided.

All necessary working should be shown on every question.

Question 5 (15 marks)

Marks

(a) Let $z = 1 - i$.

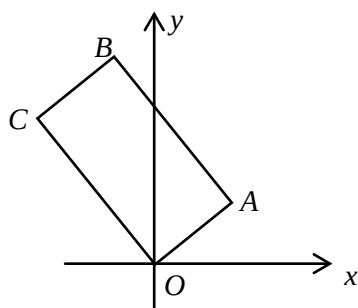
(i) Write z in modulus-argument form.

1

(ii) Hence show that $z^4 + 4 = 0$.

2

(b)



$OABC$ is a rectangle in the Argand diagram where O is the origin and the point A represents the complex number $2 + i$.

(i) Given that the length of the rectangle is three times its breadth, and OA is one of its shorter sides, find the complex numbers that correspond to B and C .

2

(ii) This rectangle is rotated about O through 30° . Find the complex number that corresponds to the new position of A .

2

(c) Consider the equation $z^3 + mz^2 + nz + 6 = 0$, where m and n are real. It is known that $1 - i$ is a root of the equation.

(i) Find the other two roots of the equation.

2

(ii) Find the values of m and n .

2

(d) If α, β and γ are the roots of the equation $x^3 - 2x^2 + 3x - 4 = 0$. Find:

(i) the equation whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$.

2

(ii) $\alpha^3 + \beta^3 + \gamma^3$.

2

Question 6 (15 marks) START A NEW PAGE**Marks**

- (a) (i) Show that $\sin x + \sin 3x = 2 \sin 2x \cos x$. **1**
- (ii) Hence, or otherwise, find all solutions of:
 $\sin x + \sin 2x + \sin 3x = 0$, for $0 \leq x \leq 2\pi$. **3**
- (b) (i) On an Argand diagram, sketch the region where the inequality
 $|z - 2i| \leq 1$ is satisfied. **2**
- (ii) Determine the maximum value of $|z|$ and $\arg z$. **2**
- (c) β is the complex root of $z^5 - 1 = 0$, with the smallest positive argument.
- (i) Show that the other complex roots of $z^5 - 1 = 0$ can be written as powers of β and plot all roots on an Argand diagram. **3**
- (ii) Find the quadratic equation whose roots are $\beta + \beta^4$ and $\beta^2 + \beta^3$. **2**
- (iii) Hence, or otherwise, deduce that $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$. **2**

End of Examination

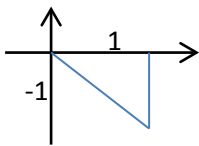
Year 11
Mathematics Extension 2
Term 4 Exam 2011
Solutions and Marking Criteria

Section I

1.	(B) $z\bar{z} = (x + iy)(x - iy)$ $= x^2 - i^2 y^2$ $= x^2 + y^2$
2.	(B) $z + iw = 3 - 4i + i(2 + i)$ $= 3 - 4i + 2i - 1$ $= 2 - 2i$
3.	(D) $\frac{z}{w} = \frac{2 + 3i}{3 - 2i} \times \frac{3 + 2i}{3 + 2i}$ $= \frac{6 + 4i + 9i - 6}{9 + 4}$ $= \frac{13i}{13}$ $= i$
4.	(A) $\tan 4\theta = 1$ $4\theta = \frac{\pi}{4} + n\pi$ $\theta = \frac{\pi}{16} + \frac{n\pi}{4}, \text{ where } n \text{ is an integer.}$

Section II

Question 5:

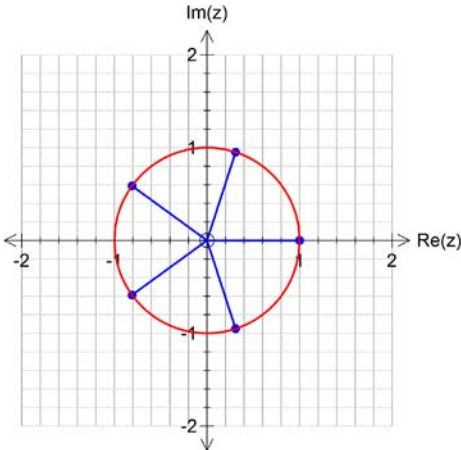
a i	 $z = \sqrt{2} \left(\cos \left(-\frac{\pi}{4} \right) + i \sin \left(-\frac{\pi}{4} \right) \right)$	1 – correct answer
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a ii	$LHS = z^4 + 4$ $= (\sqrt{2})^4 (\cos(-\pi) + i \sin(-\pi)) + 4$ $= 4(-1) + 4$ $= 0$ $= RHS$	2 – correct solution 1 – solution with one error
b i	$C = 3i(2 + i)$ $= -3 + 6i$ $B = 2 + i - 3 + 6i$ $= -1 + 7i$	2 – correct solution 1 – correct complex number for B or C only
b ii	$(2 + i)(\cos 30^\circ + i \sin 30^\circ)$ $= (2 + i)\left(\frac{\sqrt{3}}{2} + i \frac{1}{2}\right)$ $= \sqrt{3} + i + \frac{\sqrt{3}}{2}i - \frac{1}{2}$ $= \left(\sqrt{3} - \frac{1}{2}\right) + i\left(1 + \frac{\sqrt{3}}{2}\right)$	2 – correct solution 1 – correct attempt at solution
c i	$z^3 + mz^2 + nz + 6 = 0$ $1 - i \text{ is a root}$ $\therefore 1 + i \text{ is also a root}$ $\alpha(1 + i)(1 - i) = -6$ $\alpha(1 + 1) = -6$ $\alpha = -3$ $\text{Roots: } 1 \pm i, -3$	2 – correct solution 1 – correctly finds one other root
c ii	$\alpha + \beta + \gamma = -\frac{b}{a}$ $1 - i + 1 + i - 3 = -m$ $m = 1$ $\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$ $(1 + i)(1 - i) - 3(1 + i) - 3(1 - i) = n$ $1 + 1 - 3 + 3i - 3 - 3i = n$ $n = -4$	2 – correct solution 1 – correct value for m or n
d i	$x^3 - 2x^2 + 3x - 4 = 0$ $\left(\frac{1}{x}\right)^3 - 2\left(\frac{1}{x}\right)^2 + 3\left(\frac{1}{x}\right) - 4 = 0$ $\frac{1}{x^3} - \frac{2}{x^2} + \frac{3}{x} - 4 = 0$ $1 - 2x + 3x^2 - 4x^3 = 0$	2 – correct solution 1 – correct attempt at solution

d ii	α is a root of $x^3 - 2x^2 + 3x - 4 = 0$ $\therefore \alpha^3 - 2\alpha^2 + 3\alpha - 4 = 0$ $\alpha^3 = 2\alpha^2 - 3\alpha + 4$ Similarly $\beta^3 = 2\beta^2 - 3\beta + 4$ $\gamma^3 = 2\gamma^2 - 3\gamma + 4$ $\alpha^3 + \beta^3 + \gamma^3$ $= 2(\alpha^2 + \beta^2 + \gamma^2) - 3(\alpha + \beta + \gamma) + 12$ $= 2[(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)] - 3(\alpha + \beta + \gamma) + 12$ $= 2[2^2 - 2(3)] - 3(2) + 12$ $= -4 - 6 + 12$ $= 2$	2 – correct solution 1 – finding an expression for $\alpha^3 + \beta^3 + \gamma^3$
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Question 6:

a i	$LHS = \sin x + \sin 3x$ $= \sin(2x - x) + \sin(2x + x)$ $= \sin 2x \cos x - \cos 2x \sin x + \sin 2x \cos x + \cos 2x \sin x$ $= 2 \sin 2x \cos x$ $= RHS$	1 – correct solution
a ii	$\sin x + \sin 2x + \sin 3x = 0$ $2 \sin 2x \cos x + \sin 2x = 0$ $\sin 2x(2 \cos x + 1) = 0$ $\sin 2x = 0 \qquad \cos x = -\frac{1}{2}$ $2x = 0, \pi, 2\pi, 3\pi, 4\pi \qquad x = \frac{2\pi}{3}, \frac{4\pi}{3}$ $x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$ $\therefore x = 0, \frac{\pi}{2}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{3\pi}{2}, 2\pi$	3 – correct solution 2 – solution with one error 1 – showing $\sin 2x(2 \cos x + 1) = 0$
b i		2 – correct region 1 – region with one error

b ii	Max. value $z = 3$ Max. value $\arg z = \frac{\pi}{2} + \frac{\pi}{6}$ $= \frac{2\pi}{3}$	2 – correct solution 1 – correct max. value of z or $\arg z$
c i	$z^5 = 1 = \cos 0 + i \sin 0$ Let $z = r(\cos \theta + i \sin \theta)$ be the roots $\therefore z^5 = r^5(\cos 5\theta + i \sin 5\theta)$ $r^5 = 1 \quad 5\theta = 2k\pi \quad k = 0, 1, 2, 3, 4$ $r = 1 \quad \theta = \frac{2k\pi}{5}$ $z_1 = \cos 0 + i \sin 0$ $z_2 = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} = \beta$ $z_3 = \cos \frac{4\pi}{5} + i \sin \frac{4\pi}{5}$ $= \left(\cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} \right)^2$ $= \beta^2$ $z_4 = \cos \frac{6\pi}{5} + i \sin \frac{6\pi}{5} = \beta^3$ $z_5 = \cos \frac{8\pi}{5} + i \sin \frac{8\pi}{5} = \beta^4$ 	3 – correct solution 2 – correct roots but not plotted on an Argand diagram OR correct roots and plot but not in terms of powers of β. 1 – correct roots not in terms of β and not plotted on an Argand diagram

c ii	Eqn with roots $\beta + \beta^4, \beta^2 + \beta^3$ Sum of roots $= \beta + \beta^4 + \beta^2 + \beta^3$ $= -1$ (as sum of roots of $z^5 - 1 = 0$ is zero) Product of roots $= (\beta + \beta^4)(\beta^2 + \beta^3)$ $= \beta^3 + \beta^4 + \beta^6 + \beta^7$ $= \beta^3 + \beta^4 + \beta + \beta^2$ (as $\beta^5 = 1$) $= -1$ $\therefore x^2 + x - 1 = 0$	2 – correct solution 1 – progress towards correct solution
c iii	$\beta + \beta^4$ $= \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} + \cos \frac{8\pi}{5} + i \sin \frac{8\pi}{5}$ $= \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} + \cos \frac{-2\pi}{5} + i \sin \frac{-2\pi}{5}$ $= \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} + \cos \frac{2\pi}{5} - i \sin \frac{2\pi}{5}$ $= 2 \cos \frac{2\pi}{5}$ Similarly $\beta^2 + \beta^3 = 2 \cos \frac{4\pi}{5}$ $\beta + \beta^4 + \beta^2 + \beta^3 = -1$ $2 \cos \frac{2\pi}{5} + 2 \cos \frac{4\pi}{5} = -1$ $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$	2 – correct solution 1 – finding $\cos \frac{2\pi}{5}$ or $\cos \frac{4\pi}{5}$ in terms of β

Communication: (6 marks)

Question 5

- b) i) 1 mark Communication of solution
b) ii) 1 mark Communication of solution
c) i) 1 mark Communication of solution

Question 6

- a) i) 1 mark Communication of solution
c) i) 1 mark Communication of solution
c) ii) 1 mark Communication of solution